

**B5W4: Building an Amharic E-commerce Data
Extractor: Interim Submission
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1. Executive Summary & Context

EthioMart's vision is to consolidate the fragmented landscape of Ethiopian Telegram e-commerce channels into a single, seamless platform. The core business challenge addressed by this project is the lack of structured data, which hinders efficient vendor selection and the provision of micro-lending opportunities. This project directly tackles these issues by focusing on robust Named Entity Recognition (NER) extraction. Our primary objectives include sophisticated data ingestion, fine-tuning Amharic NER models for precision, conducting comprehensive model comparisons, and ultimately developing a data-driven vendor scorecard. .

2. Description of Data & Sources

The data for this project was meticulously collected from six prominent Ethiopian Telegram channels: 'ZemenExpress', 'Leyueqa', 'MerttEka', 'classybrands', 'belaclassic' and shegere. These channels were identified and connected using a custom Python scraper built with the telethon library, designed to fetch messages, images, and documents in real-time as they are posted. The raw data types primarily consist of Amharic text messages, accompanied by associated metadata such as timestamp, sender (channel title and username), and message views. While images and documents were part of the initial ingestion scope, the primary focus for NER was on the textual content.

	Channel Title	Channel Username	ID	Message	Date	Media Path
0	Zemen Express®	@ZemenExpress	6982	🔥🔥🔥.....🔥🔥🔥\n\n🔥Im...	2025-06-18 06:01:10+00:00	NaN
1	Zemen Express®	@ZemenExpress	6981	🔥🔥🔥.....🔥🔥🔥\n\n🔥B...	2025-06-16 12:21:00+00:00	NaN
2	Zemen Express®	@ZemenExpress	6980	NaN	2025-06-16 05:11:57+00:00	data/photos/@ZemenExpress_6980.jpg
3	Zemen Express®	@ZemenExpress	6979	NaN	2025-06-16 05:11:57+00:00	data/photos/@ZemenExpress_6979.jpg
4	Zemen Express®	@ZemenExpress	6978	NaN	2025-06-16 05:11:57+00:00	data/photos/@ZemenExpress_6978.jpg
...	...	...	...	...	...	...
37373	Sheger online-store	@Shageronlinestore	10	🔴 3in1 One Step Hair Dryer & Styler\n\n🔴hC...	2021-04-27 05:57:12+00:00	data/photos/@Shageronlinestore_10.jpg
37374	Sheger online-store	@Shageronlinestore	9	🟢 Home GYM - X5 slimming vibrator\n\n🔴🔴🔴🔴🔴...	2021-04-27 05:45:57+00:00	data/photos/@Shageronlinestore_9.jpg
37375	Sheger online-store	@Shageronlinestore	4	🟢Health & Personal Care\n\n🔴FingerTip Pul...	2021-04-12 08:36:40+00:00	data/photos/@Shageronlinestore_4.jpg
37376	Sheger online-store	@Shageronlinestore	3	#Finger_tip_pulse_oximeter\n\n#ብተመጣጣኝ_ዋሽ...	2021-04-12 08:35:47+00:00	data/photos/@Shageronlinestore_3.jpg
37377	Sheger online-store	@Shageronlinestore	1	NaN	2021-04-11 10:31:03+00:00	NaN
37378 rows × 6 columns						

Figure 1: collected data

Preprocessing steps were critical to transform the raw, unstructured data into a clean, unified format suitable for NER. This involved:

- **Tokenization:** Breaking down Amharic text into individual words or sub-word units. A simple Amharic tokenizer was implemented to handle the unique linguistic features of the language.
- **Normalization:** Cleaning the text by removing non-Amharic characters and extraneous whitespace, ensuring consistency across the dataset. The `clean_amharic_text` function in our script effectively handled this by retaining only Amharic characters and spaces.
- **Metadata Extraction:** Separating relevant metadata (e.g., Channel Title, Channel Username, ID, Date) from the core Message content, which was then subjected to cleaning and became the `Cleaned_Message` column.

3. Data Ingestion and Data Preprocessing

To establish a robust data ingestion system for real-time message fetching from multiple Ethiopian Telegram e-commerce channels and prepare the raw data for entity extraction. A custom scraper using the `telethon` library was developed. This scraper connected to the identified Telegram channels and collected messages, including text, images, and documents. The code was set to capture messages posted within the last two days. `telethon` library for Telegram API interaction, `csv` module for data storage, `datetime` for date handling.

- **Preprocessing Details:**
 - The `clean_amharic_text` function was applied to the 'Message' column to remove non-Amharic characters and normalize spaces, resulting in the 'Cleaned_Message' column.
 - Rows with empty 'Cleaned_Message' were removed.
 - The relevant columns (Channel Title, Channel Username, ID, Cleaned_Message, Date) were selected and saved to `cleaned_telegram_data.csv`.

```
df1=pd.read_csv('cleaned_telegram_data.csv')
```

✓ 0.1s

	Channel Title	Channel Username	ID	Cleaned_Message	Date
0	Zemen Express®	@ZemenExpress	6982	በኤሌክትሮኒክስ/ሪፖርት ለሴት መልካም መዓዛን የሚለጥ ዋጋ፡- ብር ወሰን ፍሬ ነ...	2025-06-18 06:01:10+00:00
1	Zemen Express®	@ZemenExpress	6981	በፈለጉት አቅጣጫ ልጅዎን በምቸት ማዘል ያስችልዎታል ዋጋ፡- ብር ወሰን ፍሬ...	2025-06-16 12:21:00+00:00
2	Zemen Express®	@ZemenExpress	6973	በኤሌክትሮኒክስ የሚለራ ለሴትና ለመኪና መልካም መዓዛን የሚለጥ ዋጋ፡- ብር ወ...	2025-06-16 05:11:57+00:00
3	Zemen Express®	@ZemenExpress	6972	በኤሌክትሮኒክስ የሚለራ ለሴትና ለመኪና መልካም መዓዛን የሚለጥ ዋጋ፡- ብር ወ...	2025-06-16 05:11:26+00:00
4	Zemen Express®	@ZemenExpress	6970	ዋጋ፡- ብር ወሰን ፍሬ ነው ያለን እይራሻ መገናኛመስረትደፋርዋልሁተኛፎቅ ...	2025-06-16 05:09:03+00:00
...	...	...	...	...	...
21434	Sheger online-store	@Shageronlinestore	12	ለኪችንዎ ውበት እጅግ ተመራጭ ውሀ የሚያስገባ ቅባት ዘይት ነገሮች የሚያስገቡ...	2021-04-27 05:58:59+00:00
21435	Sheger online-store	@Shageronlinestore	10	ከርል ለመስራት ለማለለስ እንዲሁም ለማድረቅ የሚያስገባል ለአትየቶንዎን...	2021-04-27 05:57:12+00:00
21436	Sheger online-store	@Shageronlinestore	9	ታላቅ ቅናሽ የሰጧትዎ ውፍረት እስከሚታል ውሃ በውሃ በርዋን በእጭር ጊዜ...	2021-04-27 05:45:57+00:00
21437	Sheger online-store	@Shageronlinestore	4	ለጤናችን ዋጋ፡- ብር ያለበት ደረሰ በቶ እናደርገለን	2021-04-12 08:36:40+00:00
21438	Sheger online-store	@Shageronlinestore	3	በተመጣጣኝዎታ ለአጠቃቀም ምቹ በሰጧትችን ያለውን የኦክስጅን እና የልብ ...	2021-04-12 08:35:47+00:00

21439 rows × 5 columns

Figure 2: Preprocessed data

4. Label a Subset of Dataset in CoNLL Format

The dataset was labeled in the CoNLL format to identify and label entities such as products, prices, and locations in Amharic text. The labeling process involved user interaction to classify tokens, ensuring accurate entity recognition for subsequent NER model training.

```
sentence
✓ 0.0s

[( 'በኤፌክትሪክየሚሰራ', 'O'),
  ('ለቤት', 'O'),
  ('መልካም', 'O'),
  ('መዓዛን', 'B-PRODUCT'),
  ('የሚሰጥ', 'I-PRODUCT'),
  ('ዋጋ፡-', 'B-LOC'),
  ('ብር', 'I-LOC'),
  ('ውስን', 'B-PRICE'),
  ('ፍሬ', 'I-PRICE'),
  ('ነው', 'O'),
  ('ያለን', 'B-PRODUCT'),
  ('እድራሻ', 'I-PRODUCT'),
  ('መገናኛመሰረትደፋርሞልሁለተኛፎቅ', 'B-LOC'),
  ('ቢሮ', 'I-LOC'),
  ('ቁ', 'B-PRICE'),
  ('በ', 'I-PRICE'),
  ('ለማዘዝ', 'O'),
  ('ይጠቀሙ', 'B-PRODUCT'),
  ('ለተጨማሪ', 'I-PRODUCT'),
  ('ማብራሪያ', 'B-LOC'),
  ('የቴሌግራም', 'I-LOC'),
  ('ገፃችን', 'B-PRICE')]
```

Figure 3: Labeled susbset of data

In conclusion, the project aims to foster a unified e-commerce experience by leveraging advanced data processing techniques, thereby enhancing vendor selection and micro-lending opportunities in the Ethiopian market.